

# Does Retail Attention Predict Returns? Evidence from Robinhood “Popularity” Data, 2018–2020

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## Abstract

A large body of behavioral finance research, beginning with Barber and Odean (2000), finds that individual investors who trade frequently tend to underperform the market, partly due to their tendency to chase recently attention-grabbing stocks rather than making decisions based on new information. Commission-free, “gamified” trading apps such as Robinhood have been argued to amplify this tendency by directing user attention toward stocks with extremely recent price moves. Using historical data on Robinhood user holdings (“popularity”), along with daily stock prices for 57 stocks from May 2018 to August 2020, this paper examines whether increases in retail popularity predict subsequent returns. The results indicate that a stock’s popularity change relative to its own history does not significantly predict its return over the following week. However, when attention is measured relative to other stocks on the same day, identifying stocks likely to appear on a “Top Movers”-style list, stocks in the top 5% of daily popularity change underperform the rest of the sample by about 0.95 percentage points over the following week ( $p = 0.007$ ). These results suggest that attention-driven underperformance depends on relative attention rather than absolute popularity changes and that specification choices substantially affect whether this effect is detectable.

## 1 Introduction

The 2010s and early 2020s brought a drastic shift in who trades stocks and how. For most of the preceding decades, initiating a trade meant going through a stock broker and paying a commission of a large percent of the trade per transaction, a cost that discouraged frequent, smaller trades and kept most of the trading concentrated among more dedicated, usually wealthier investors. Beginning around 2013 and continuing through the late 2010s and 2020s, commissions fell to zero, brokerages became easy to access on smartphones, and mobile apps featured game-like elements like “Top Movers” lists and confetti animations on completed trades. These changes lowered the barriers

to trading for millions of new, inexperienced investors. Robinhood became the most prominent symbol of this shift.

This paper asks one small question of a much larger debate about whether attention-driven trading is worth it for individual investors. Barber and Odean showed that frequent trading by individual investors tends to underperform a simple buy-and-hold strategy, and that overconfidence is a large reason why investors trade as much as they do. More recent research specifically on Robinhood claims that the app’s design pushes users toward stocks that actually underperform.

I test this using a metric of Robinhood that, unusually, was made public for several years: an aggregate count of how many Robinhood accounts held each stock, updated hourly. A third-party project called Robintrack archived this data from May 2018 until Robinhood discontinued this public feed in August 2020. By combining this popularity data with daily stock prices, I can ask directly: when a stock’s Robinhood popularity increases sharply, what happens to its price afterward?

## 2 Literature Review

The starting point for this literature is Barber and Odean’s “Trading Is Hazardous to Your Wealth” (2000), which used brokerage records for over 66,000 households between 1991 and 1996. The most active traders in this sample earned substantially lower annual returns than the market, and lower returns than less active investors in the same sample, even before accounting for the additional costs of trading more frequently mentioned previously. The authors attributed this effect primarily to overconfidence. Investors who believe they have private information trade more often than is rational, and on average this additional trading reduces rather than increases returns.

A related paper, “Boys Will Be Boys” (Barber and Odean, 2001), used the same data to show that this pattern is more pronounced for men than women, consistent with psychological evidence that men tend to be more overconfident when making financial decisions. Men in their sample traded about 45% more than women, and as a result, earned lower risk-adjusted returns, with the gap larger among single investors. This paper is useful less for its specific finding on gender and more as an example of a research design: take an established prediction and test it on a subgroup where the underlying mechanism should be stronger or weaker.

Coval, Hirshleifer, and Shumway, using the same brokerage data, add an important qualification: not every individual investor underperforms. The top quarter of accounts in the Barber–Odean sample actually outperformed the market by a small margin. This raises the question of whether some traders have genuine skill or whether this is just the variance one would expect given how undiversified many small, individual portfolios are. Some individuals may end up on the lucky side of the distribution due to chance alone.

The data that this literature is based on is from the 1990s, a period of mostly phone-based trading. The relevant question today is whether this same pattern of underperformance holds or is magnified in an environment of free, app-based trading with design features intended to increase engagement and attention. Barber, Huang, Odean, and Schwarz (2022) tackle this using Robinhood data. They find that Robinhood’s “Top Movers” list—the twenty stocks with the largest price moves on a given day—has a measurable effect on user behavior. In fact, roughly 35% of Robinhood users’

net buying activity in this sample went into stocks on this list. Periods of heavy buying driven by this attention tend to correlate with negative returns afterward.

A separate line of research attempts to isolate the effect of the design of trading apps themselves, using experiments. Zoican, Chapkovski, and Khapko ran a randomized experiment in which participants traded using either a plain interface or one with “gamified” features such as animations, with both groups given the same information about the stocks. The gamified interface led participants to trade more and take on riskier positions, showing that design choices alone, aside from the actual users, can change how people trade.

Together, this research suggests the following sequence: (1) frequent, attention-driven trading by individuals tends to underperform; (2) modern trading apps direct attention toward a small set of “spotlight” stocks; and (3) being in that spotlight predicts worse subsequent returns for the stock. This paper tests the third step directly, using a broader sample of stocks than the original Top Movers study, and compares two different measures of “attention spike” to assess how sensitive the result is to that choice.

## 3 Data

### 3.1 Sources

Robinhood popularity data comes from Robintrack, a project that regularly recorded the number of Robinhood accounts holding each stock from Robinhood’s public data feed. This data is available from May 2, 2018 through August 13, 2020, when Robinhood discontinued public access. Daily adjusted closing prices come from Yahoo Finance.

### 3.2 Sample

The sample consists of 57 stocks selected to span a range of retail attention during this period. It includes large, widely held companies (Apple, Microsoft, Amazon, Walmart, Coca-Cola) as well as stocks closely associated with the 2020–2021 retail trading boom (GameStop, AMC, Nikola, Tilray, Plug Power, Nio, DraftKings, Peloton, Zoom). For some analyses, the sample is divided into a “high-attention” group of 19 stocks frequently described as “meme” or retail-favorite stocks during this period, and the remaining 38 stocks.

### 3.3 Variable Construction

For each stock  $i$  and trading day  $t$ , let  $H_{i,t}$  denote the number of Robinhood accounts holding stock  $i$  on day  $t$ , and  $P_{i,t}$  denote its adjusted closing price. I construct the following variables.

**Popularity change (own-history measure).** For a window of  $k$  trading days ( $k \in \{1, 5\}$ ), the popularity change is the log difference in holders:

$$\text{PopLogChg}_{i,t}^{(k)} = \ln(H_{i,t}) - \ln(H_{i,t-k}) \quad (1)$$

A log difference is used, rather than a simple percent change, because it is symmetric around zero and behaves better given the very large swings in holder counts that some smaller stocks experienced.

**Forward return (the outcome variable).** Over the same horizon  $k$ , the forward return is the stock’s simple percentage price change over the following  $k$  days:

$$\text{FwdRet}_{i,t}^{(k)} = \frac{P_{i,t+k}}{P_{i,t}} - 1 \quad (2)$$

Note the timing:  $\text{PopLogChg}_{i,t}^{(k)}$  uses information available up to and including day  $t$ , while  $\text{FwdRet}_{i,t}^{(k)}$  is realized strictly after day  $t$ . This ordering is what makes the regression a test of whether attention predicts future returns, rather than a description of a contemporaneous correlation.

**Cross-sectional attention rank (Top Mover proxy).** For each day  $t$ , I rank all 57 stocks by their 1-day popularity change and convert the rank to a percentile:

$$\text{Rank}_{i,t} = \frac{\#\{j : \text{PopLogChg}_{j,t}^{(1)} \leq \text{PopLogChg}_{i,t}^{(1)}\}}{57} \quad (3)$$

A stock is flagged as having an “extreme attention” day if it falls in the top 5% of the cross-section that day:

$$\text{ExtremeAttention}_{i,t} = \mathbb{1}[\text{Rank}_{i,t} \geq 0.95] \quad (4)$$

That is,  $\text{ExtremeAttention}_{i,t} = 1$  if stock  $i$ ’s one-day popularity increase was among the largest 5% of all 57 stocks’ popularity changes on that specific day, and 0 otherwise.

After merging popularity and price data and trimming the top and bottom 0.5% of observations on each variable (to limit the influence of data artifacts, such as days when Robintrack’s scrape produced an implausible jump in the holdings count), the main panel contains 28,107 stock-day observations across all 57 tickers.

## 4 Methodology

I estimate three sets of specifications, all variants of ordinary least squares (OLS).

**Specification 1 (own-history, 5-day).** The baseline pooled regression is:

$$\text{FwdRet}_{i,t}^{(5)} = \beta_0 + \beta_1 \cdot \text{PopLogChg}_{i,t}^{(5)} + \varepsilon_{i,t} \quad (5)$$

I also estimate a version that includes stock fixed effects,  $\alpha_i$ , which control for permanent differences across stocks (for example, some stocks are simply more volatile than others on average), and identify  $\beta_1$  from variation within each stock over time:

$$\text{FwdRet}_{i,t}^{(5)} = \alpha_i + \beta_1 \cdot \text{PopLogChg}_{i,t}^{(5)} + \varepsilon_{i,t} \quad (6)$$

In both versions,  $\beta_1$  is the coefficient of interest. It measures how much higher or lower a stock’s return over the next five days is, on average, when its popularity grows faster than usual over the prior five days. Standard errors are clustered by stock in all specifications, which accounts for the fact that observations from the same stock are not independent of one another. For example, the five-day window ending on a Monday and the one ending on the following Tuesday share four days of data.

I also report a non-parametric version of Specification 1: stocks are sorted into five groups (quintiles) based on  $\text{PopLogChg}_{i,t}^{(5)}$ , and average forward returns are compared across groups. This approach does not assume the relationship is linear.

**Specification 2 (own-history, 1-day).** Identical in form to Specification 1, but with  $k = 1$  throughout:

$$\text{FwdRet}_{i,t}^{(1)} = \beta_0 + \beta_1 \cdot \text{PopLogChg}_{i,t}^{(1)} + \varepsilon_{i,t} \quad (7)$$

**Specification 3 (cross-sectional “Top Mover” proxy).** Here the explanatory variable is the binary  $\text{ExtremeAttention}_{i,t}$  indicator defined in Section 3.3:

$$\text{FwdRet}_{i,t}^{(5)} = \beta_0 + \beta_1 \cdot \text{ExtremeAttention}_{i,t} + \varepsilon_{i,t} \quad (8)$$

Because the explanatory variable is a binary indicator, this specification has a direct interpretation:  $\beta_0$  is the average forward return on “normal” days, and  $\beta_0 + \beta_1$  is the average forward return on “Top Mover” days. The coefficient  $\beta_1$  is exactly the difference between the two group means reported in Table 2; the regression and the simple comparison of means are equivalent, with the regression additionally providing a  $p$ -value.

## 5 Results

### 5.1 Specification 1: Own-History, 5-Day Window

Table 1 reports the average 5-day forward return by quintile of 5-day popularity change.

Table 1: Average forward 5-day return by popularity-change quintile (own-history)

| Quintile (1 = largest popularity decrease, 5 = largest increase) | Avg. forward return | $N$   |
|------------------------------------------------------------------|---------------------|-------|
| 1                                                                | 0.26%               | 5,622 |
| 2                                                                | 0.77%               | 5,621 |
| 3                                                                | 0.67%               | 5,621 |
| 4                                                                | 0.70%               | 5,621 |
| 5                                                                | 0.76%               | 5,622 |

There is no monotonic pattern across quintiles, and the difference between the top and bottom quintiles (0.50 percentage points) is small relative to the standard deviation of returns. The pooled regression confirms this. Substituting the estimated coefficients into Equation (5):

$$\widehat{\text{FwdRet}}_{i,t}^{(5)} = \underbrace{0.0059}_{\hat{\beta}_0} + \underbrace{0.0230}_{\hat{\beta}_1} \cdot \text{PopLogChg}_{i,t}^{(5)}$$

with  $p = 0.391$  on  $\hat{\beta}_1$ , indicating no significant relationship. Adding stock fixed effects (Equation (6)) reduces  $\hat{\beta}_1$  to 0.0104 ( $p = 0.712$ ).  $R^2$  is below 0.02 in both cases. By this measure, an increase in a stock's popularity relative to its own history provides no information about its subsequent return.

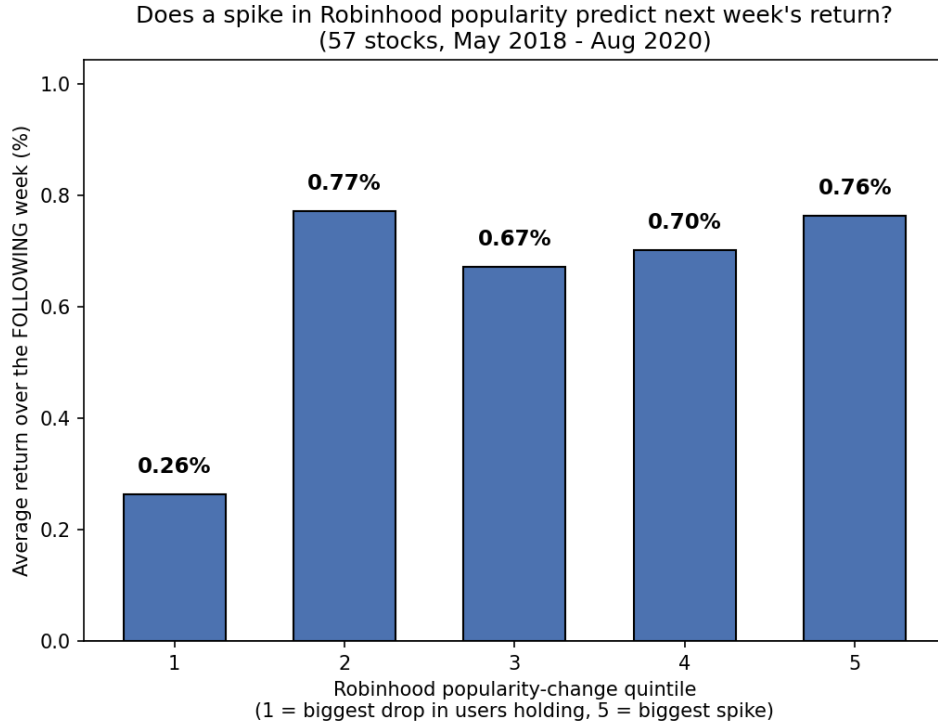


Figure 1: Average return over the following week, by quintile of 5-day Robinhood popularity change (57 stocks, May 2018–Aug 2020).

## 5.2 Specification 2: Own-History, 1-Day Window

Using a one-day window does not change this result:

$$\widehat{\text{FwdRet}}_{i,t}^{(1)} = \underbrace{0.0011}_{\hat{\beta}_0} - \underbrace{0.0029}_{\hat{\beta}_1} \cdot \text{PopLogChg}_{i,t}^{(1)}, \quad p = 0.911$$

and the quintile averages (0.10% to 0.16%) show no meaningful difference.

### 5.3 Specification 3: Cross-Sectional “Top Mover” Proxy

The result changes substantially when attention is measured relative to other stocks on the same day rather than relative to a stock’s own history. Approximately 5.8% of stock-day observations (1,674 of 28,617) fall in the top 5% of same-day popularity changes. Table 2 compares average forward returns for these “Top Mover” days with all other days.

Table 2: Average forward 5-day return: “Top Mover” days vs. other days

| Group                     | Avg. forward return | $N$    |
|---------------------------|---------------------|--------|
| Other days (bottom 95%)   | 0.68%               | 26,943 |
| “Top Mover” days (top 5%) | −0.27%              | 1,674  |

The difference, −0.95 percentage points, is statistically significant ( $p = 0.007$ ). In the notation of Section 4:

$$\widehat{\text{FwdRet}}_{i,t}^{(5)} = \underbrace{0.0068}_{\hat{\beta}_0} - \underbrace{0.0095}_{\hat{\beta}_1} \cdot \text{ExtremeAttention}_{i,t}, \quad p = 0.007, \quad R^2 = 0.0009$$

so  $\hat{\beta}_0 = 0.68\%$  is the average return following a “normal” day, and  $\hat{\beta}_0 + \hat{\beta}_1 = 0.68\% - 0.95\% = -0.27\%$  is the average return following a “Top Mover” day, matching the values in Table 2. Stocks with an extreme one-day increase in Robinhood popularity relative to other stocks underperform over the following week.

Restricting the sample to the 19 “high-attention” stocks produces a similar point estimate (−1.02 percentage points) that is no longer statistically significant ( $p = 0.120$ ), likely due to the smaller sample size (749 “Top Mover” observations among meme stocks, compared with 1,674 in the full sample) and the higher volatility of these stocks.

### 5.4 Case Study: GameStop

Although the Robintrack data ends in August 2020, five months before GameStop’s short squeeze, the popularity series for GME over this period remains informative. Figure 3 shows GameStop’s adjusted closing price alongside the number of Robinhood accounts holding the stock from mid-2018 through August 2020.

Robinhood holdings of GME increased roughly threefold, from approximately 7,000 to over 30,000 accounts, between early 2019 and mid-2020, a period during which the stock’s price was flat or declining. This is consistent with the broader finding that retail attention does not reliably track near-term price performance, and illustrates that the retail interest that became prominent in January 2021 had been building for over a year beforehand.

## 6 Discussion

The central finding of this paper is that whether an “attention predicts underperformance” effect can be detected depends on how retail attention is measured. When a stock’s popularity is measured

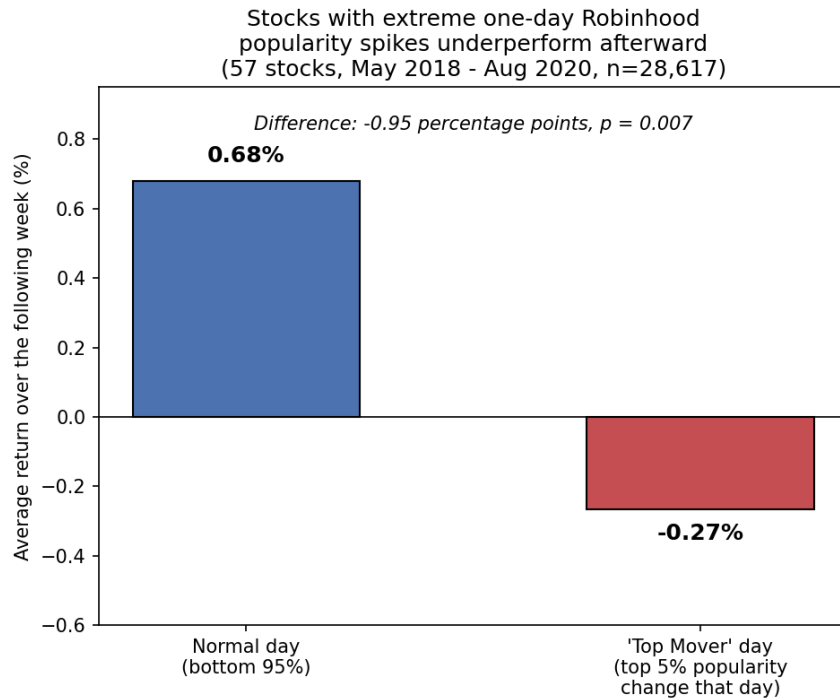


Figure 2: Average return over the following week, comparing “Top Mover” days (top 5% of cross-sectional popularity change) to all other days.

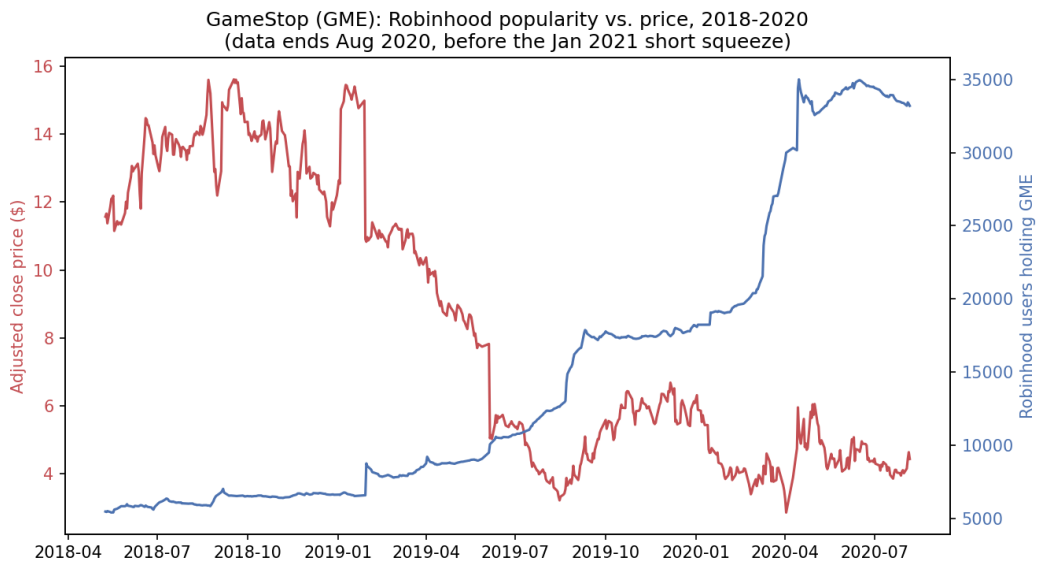


Figure 3: GameStop (GME): Robinhood popularity vs. adjusted closing price, 2018–2020.



against its own history—that is, whether more of its existing holders are buying than usual—it shows no detectable relationship with subsequent returns. But when popularity is measured against other stocks on the same day, whether the stock stands out as unusually popular relative to the rest of the market, it does predict underperformance, by roughly 1 percentage point over the following week.

This distinction lines up with the mechanism Barber et al. (2022) describe. Robinhood’s Top Movers list is inherently a relative ranking. It selects twenty stocks from thousands based on how their price moves compared with the rest of the market that day. A stock can experience an unusually large move relative to its own history without being unusual enough to appear on a list of the most extreme movers across the entire market. Specification 1 may therefore capture a different phenomenon, such as ordinary mean-reversion or momentum within a stock. Specification 3 is a closer proxy for the spotlight mechanism itself.

It is also worth noting what this paper does not show. The effect identified in Specification 3, while statistically significant, is modest: a difference of roughly 1 percentage point over a week is unlikely to exceed standard trading costs, and identifying “Top Mover” days in real time, before other market participants react, would be difficult in practice. The result is therefore better interpreted as evidence that retail attention is meaningful in some forms and not others rather than a trading strategy.

## 7 Limitations and Future Research

Several limitations should be noted. First, the sample period (May 2018–August 2020) includes the COVID-19 market decline and one of the fastest recoveries on record; this period may overstate or understate attention effects relative to more typical market conditions. Second, the Robintrack data ends in August 2020 and therefore cannot speak to the GameStop short squeeze of January 2021 or the broader “meme stock” episodes that followed, which are arguably the clearest examples of the phenomenon examined here. Third, the “extreme attention” measure in Specification 3 is a simplified proxy: Robinhood’s actual Top Movers list is based on price movements rather than popularity changes, although the two are likely correlated. Future work could construct a more direct measure using daily price-movement rankings, or extend the analysis using attention data from other platforms, such as mention frequency on Reddit’s WallStreetBets, that remains available after 2020.

## 8 Conclusion

Using historical Robinhood popularity data combined with stock prices for 57 stocks from 2018 to 2020, this paper finds no evidence that a stock’s popularity change relative to its own history predicts its subsequent return. However, stocks experiencing an extreme popularity increase relative to other stocks on the same day—a proxy for appearing on a “Top Movers” list—underperform the rest of the sample by approximately 1 percentage point over the following week, a statistically significant result consistent with the finding that Robinhood’s attention-directing features channel buying into stocks that subsequently underperform. The contrast between these specifications sug-

gests that the underperformance documented in prior research reflects relative attention specifically, and that conclusions based on a stock’s own history may not generalize to its standing relative to the broader market at a given time.

## References

Barber, B. M., & Odean, T. (2000). Trading is hazardous to your wealth: The common stock investment performance of individual investors. *The Journal of Finance*, 55(2), 773–806.

Barber, B. M., & Odean, T. (2001). Boys will be boys: Gender, overconfidence, and common stock investment. *The Quarterly Journal of Economics*, 116(1), 261–292.

Barber, B. M., Huang, X., Odean, T., & Schwarz, C. (2022). Attention-induced trading and returns: Evidence from Robinhood users. *The Journal of Finance*.

Coval, J. D., Hirshleifer, D. A., & Shumway, T. (2005). Can individual investors beat the market? Working paper, Harvard Business School.

Zoican, M., Chapkovski, P., & Khapko, M. (2023). How gamification affects investor behavior. University of Toronto working paper.

## A Data and Code Appendix

All code used to construct the dataset and run the analyses in this paper is included in the accompanying replication files: `build_panel.py` (constructs the Specification 1/2 panel), `alt_specs.py` (Specifications 2 and 3), `run_analysis.py` (regression tables), and `make_charts.py` / `make_extreme_chart.py` (figures). The underlying popularity data was sourced from an archived copy of the Robintrack dataset; price data from Yahoo Finance.